East West University
Department of Computer Science and Engineering

Course: CSE 475 (Machine Learning)
Section - 03

Submitted To :	**Submitted By :**
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Lab Report:06

Result of Different Models: The models achieved accuracies ranging from 80% to 100%, indicating that the dataset is highly predictable. XGBoost and Decision Tree produced the highest accuracy (100%), while KNN had the lowest performance (80%) due to its sensitivity to the data distribution.

```
Random Forest 0.95
XGBoost 1.0
AdaBoost 0.95
KNN 0.8
SVM 0.9
Decision Tree 1.0
```

Figure : Accuracy of Models

Model	Accuracy
XGBoost	1.00
Decision Tree	1.00
Blending	1.00
Stacking	1.00
Soft Voting	1.00
Hard Voting	1.00
Random Forest	0.95
AdaBoost	0.95
SVM	0.90
KNN	0.80

Figure : . Accuracy Comparison of Individual and Ensemble

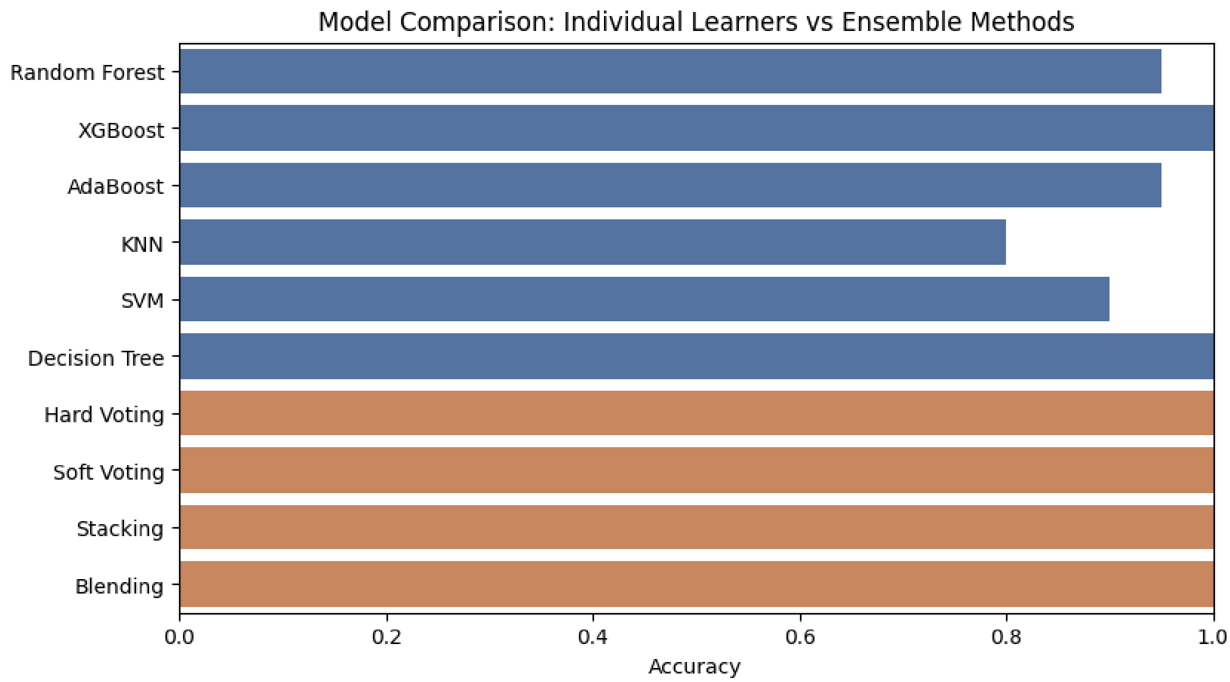

Figure: Model Comparison

Conclusion: Among the individual models, XGBoost and Decision Tree achieved the highest accuracy of 100%, while Random Forest and AdaBoost achieved 95%, SVM achieved 90%, and KNN achieved 80%. Among the ensemble methods, Hard Voting, Soft Voting, Stacking, and Blending also achieved 100% accuracy, indicating that combining multiple models improved the overall classification performance on this dataset.

Code : Ensemble.ipynb